SMART HAT

BY: CIRCUIT BREAKERS

"NEXT-GENERATION SAFETY FOR CONSTRUCTION AND INDUSTRIAL WORKERS"

MEET THE TEAM

Landon Moore, EE

Matthew Dykes, CPE

Aaron Ramsay, EE

Dawson Carnathan, EE

Leigh Kergosien, EE

Junming Diao, Advisor

PROBLEM STATEMENT

- Life-threatening hazards in industrial and construction fields.
- Lack of feedback on dangers and injuries can delay timely help for workers.
- In 2023, there were 5283 fatal work-related injuries in the US alone[1].

TARGET AUDIENCE

Construction Companies and Contractors
(general/subcontractors, supervisors)

Manufacturing and Industrial Plants
(factory and assembly workers)

Mining Operations
(pipeline, underground, rig, and mining crews)

Regulatory Agencies
(OSHA and safety boards)

Infrastructure Companies
(maintenance / installation crews)

[2]

REAL-WORLD EXAMPLES

Location: Stamford, Connecticut

- Worker at Rubino Brothers Inc. was found unconscious on scaffolding, after being struck in the temple by an object, while wearing all necessary safety equipment.
- No workers witnessed the accident[3].

Location: Yale University

- 14 workers were hospitalized due to high carbon monoxide levels around a construction site.
- Ventilation was on site. However, there was still CO buildup in the air[4].

[4]

SOLUTION: SMART HAT

- Immediate Accident Response
- Real-time Vitals
- Hazard Detection + Prevention

SUBSYSTEMS

Chemical Detection
Dawson Carnathan
Leigh Kergosien

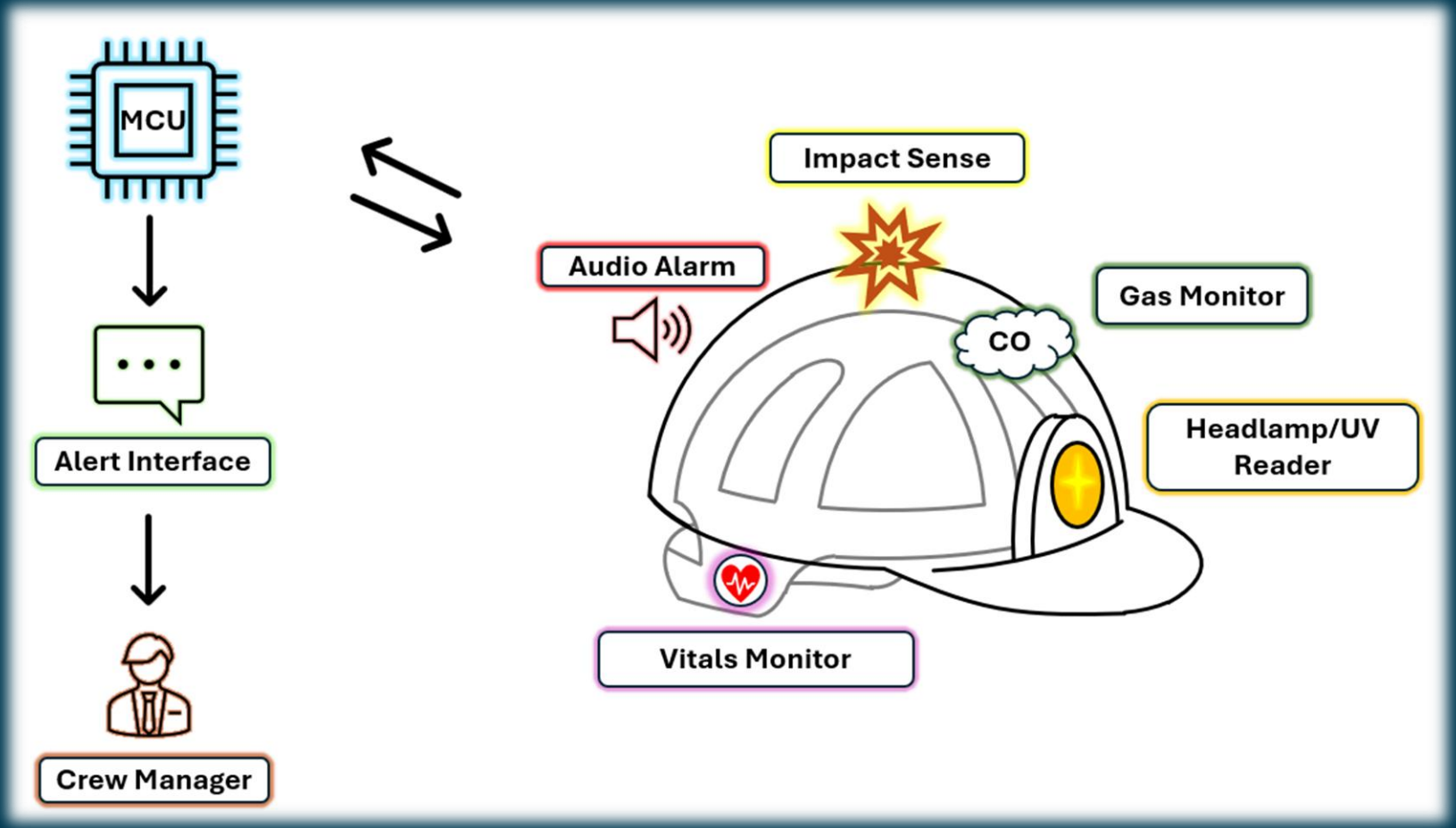
Headlamp & UV Monitoring
Aaron Ramsay
Matthew Dykes

Impact Detection
Leigh Kergosien
Landon Moore

Vitals Monitoring
Landon Moore
Dawson Carnathan

SMS/App Communication
Matthew Dykes
Aaron Ramsay

HIGH LEVEL DESCRIPTION



COMPETITIVE ADVANTAGE

SMART HAT

- Multiple capabilities all in one
- Impact detection
- CO detection
- Real time alerts
- Vitals monitoring

GUARDIO

- Real time alerts
- Impact/Fall detection only
- Not yet available to public

[5]

BROADER IMPACT

1. Companies can improve worker well-being and safety culture.
2. Workplace accidents will decrease with improved communication, collaboration, and real time awareness.
3. The same smart PPE technology can be adapted for medical responders, chemical plants, and many other hazardous environments.

[6]

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THANK YOU